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How Much Information Remains Under Censoring? A Variance Decomposition for the Kaplan–Meier Estimator

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ABSTRACT

The Kaplan–Meier estimator is standard for censored survival data, but information lost to censoring is rarely made explicit. The practical question is not only how many subjects remain at risk, but also how much information remains for inference. We develop a variance-decomposition framework that makes this loss explicit by separating complete-follow-up variability from the additional component induced by censoring. This yields pointwise measures of information retention: a variance-inflation factor, relative efficiency, and an effective sample size, $\widehat{ESS}(t)$. $\widehat{ESS}(t)$ has a binomial-equivalent interpretation: it is the size of a hypothetical complete-follow-up binomial experiment with survival probability $\widehat{S}_{KM}(t)$ whose variance equals the Greenwood variance. The framework also yields a pointwise censoring-component interval, showing where the survival estimate from the same sample would plausibly have fallen if censoring had not occurred. Simulations show that $\widehat{ESS}(t)$ is well calibrated and that the censoring-component interval has near-nominal pointwise coverage. Examples from clinical trials illustrate that risk-set size and $\widehat{ESS}(t)$ answer different questions: a small number at risk does not imply low precision when risk-set depletion is driven primarily by observed events rather than censoring. The proposed tools provide practical data-maturity summaries for judging how much information remains over time and how reliable later Kaplan–Meier inference is.

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1. Introduction

The Kaplan–Meier estimator (Kaplan and Meier 1958) remains the standard method for summarizing survival data when some participants are censored. Its large-sample properties have long been established (Breslow and Crowley 1974; Meier 1975; Gill 1980; Stute 1995), yet the distinct contribution of censoring to the estimator’s variability—though it has been addressed in the theoretical literature, has received little attention in applied research and is seldom made explicit in published analyses. Censoring inevitably leads to information loss, but this loss is rarely quantified in practice. Conventional summaries, such as the number at risk, show how many participants remain under observation but say little about how much statistical information those observations still contain. This paper revisits the structure of the Kaplan–Meier estimator to make that loss explicit. By decomposing the variance of the Kaplan–Meier estimator, we separate the variability that would exist under complete observation from the additional uncertainty introduced by censoring. From this perspective, censoring can be viewed as inflating variance or, equivalently, reducing information. The resulting measures of information retention and effective sample size provide an intuitive, variance-based view of how much information remains at each time point. The theoretical foundation for this development is the classical representation of the

Kaplan–Meier estimator due to Peterson Jr (1977) and the corresponding influence-function decomposition of Reid (1981). Building on these results, and on earlier work on information loss and relative efficiency under censoring (Oakes 1977; Benedetti et al. 1982; Hackmann et al. 2025), the contribution of this paper is to convert the classical decomposition into a practical information accounting framework for Kaplan–Meier curves. The framework separates the complete-data component of the Kaplan–Meier variance from the additional component induced by censoring. This separation leads directly to pointwise measures of information retention, including the variance-inflation factor, relative efficiency, and effective sample size. It also provides a pointwise interval for the latent complete-data survival estimate, characterizes the attenuation of correlation under censoring, and interprets effective sample size as the size of a variance-equivalent complete-follow-up binomial experiment. This binomial interpretation links effective sample size to ordinary binomial precision, to finite-sample normal approximation through Berry–Esseen bounds, and to the accumulated risk-set history underlying Greenwood’s variance.

Applied to both simulated and real clinical trial data, these measures show that small risk sets need not imply weak statistical information. They provide a variance-based way to assess information loss under censoring and to interpret Kaplan–Meier

curves in terms of the precision they retain. The following sections formalize the decomposition, develop the resulting information measures, and illustrate their interpretation through simulation and clinical trial examples.

2. Latent Survival, Kaplan–Meier, and the Censoring-Noise Decomposition

Let X denote the event time and C an independent right-censoring time; observe $T = \min(X, C)$ and $\delta = \mathbf{1}\{X \leq C\}$, with $X \perp C$, $X \sim F$, $C \sim G$. The survival function of interest is

$$S(t) = \Pr(X > t), \quad t \geq 0. \quad (1)$$

If all event times were observed, the empirical survival function would simply be

$$\begin{aligned} \tilde{S}_n(t) &= \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i > t\}, & E[\tilde{S}_n(t)] &= S(t), \\ \text{var}(\tilde{S}_n(t)) &= \frac{S(t)\{1 - S(t)\}}{n}. \end{aligned} \quad (2)$$

To describe how the observed sample evolves over time, let

$$N(t) = \sum_{i=1}^n \mathbf{1}\{T_i \leq t, \delta_i = 1\}, \quad N^c(t) = \sum_{i=1}^n \mathbf{1}\{T_i \leq t, \delta_i = 0\}, \quad (3)$$

denote the cumulative numbers of observed events and censored observations by time t , and let $R(t)$ denote the number at risk just prior to time t . Then

$$R(t) = n - N(t-) - N^c(t-), \quad t \geq 0, \quad (4)$$

so that, at time $t-$, each subject is either still at risk, has failed, or has been censored. Under censoring, $\tilde{S}_n(t)$ is latent, and the Kaplan–Meier estimator replaces it using the observed event times. Assuming no ties, let $t_1 < \dots < t_m$ denote the distinct observed event times. Then

$$\hat{S}_{KM}(t) = \prod_{t_j \leq t} \left(1 - \frac{\delta_j}{R(t_j)}\right). \quad (5)$$

Consequently $\hat{S}_{KM}(t)$ is defined only at the observed event times and is piecewise constant between them, so the estimate (and its associated variance estimate) is carried forward unchanged until the next event time. Greenwood's estimator gives the associated variance:

$$\begin{aligned} \hat{v}_{KM}(t) &= \hat{S}_{KM}^2(t) \hat{\sigma}^2(t), \\ \hat{\sigma}^2(t) &= \sum_{t_j \leq t} \frac{\delta_j}{R(t_j)\{R(t_j) - \delta_j\}}. \end{aligned} \quad (6)$$

Both $\hat{S}_{KM}(t)$ and $\tilde{S}_n(t)$ are asymptotically linear for fixed t , with influence functions

$$\sqrt{n}(\tilde{S}_n(t) - S(t)) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}_{\tilde{S}}(X_i; t) + o_p(1), \quad (7)$$

$$\sqrt{n}(\hat{S}_{KM}(t) - S(t)) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}_{KM}(X_i, \delta_i; t) + o_p(1), \quad (8)$$

and the orthogonal decomposition (Peterson/Reid)

$$\begin{aligned} \text{IF}_{KM}(X, \delta; t) &= \underbrace{\{\mathbf{1}(X > t) - S(t)\}}_{\text{IF}_{\tilde{S}}(X; t)} + Z_t(X, \delta), \\ E[Z_t] &= 0, \quad E[Z_t \text{IF}_{\tilde{S}}(X; t)] = 0. \end{aligned} \quad (9)$$

This decomposition goes back to Peterson Jr (1977), who expressed the Kaplan–Meier curve via two sub-survival functions $S_1^*(t) = \Pr(T > t, \delta = 1)$ and $S_2^*(t) = \Pr(T > t, \delta = 0)$ under independent censoring. Reid (1981) introduced the corresponding influence-function representation: IF_{KM} splits into the complete-data empirical component $\text{IF}_{\tilde{S}}$ (driven by events, S_1^*) and an orthogonal mean-zero term Z_t reflecting the censoring contribution (S_2^*).

Taking second moments yields

$$v_{KM}(t) \approx \frac{1}{n} E[\text{IF}_{KM}^2], \quad v_{\tilde{S}}(t) \approx \frac{1}{n} E[\text{IF}_{\tilde{S}}^2] = \frac{S(t)\{1 - S(t)\}}{n}. \quad (10)$$

Using (9) and $E[Z_t \text{IF}_{\tilde{S}}] = 0$, the Kaplan–Meier variance can be expressed as the sum of the latent complete-data variance and an additive term capturing the variability introduced by censoring

$$v_{KM}(t) = v_{\tilde{S}}(t) + \frac{1}{n} \text{var}\{Z_t(X, \delta)\}. \quad (11)$$

For the next result, we assume the standard regularity conditions for Kaplan–Meier asymptotics. The observations are iid; censoring is independent of the event time, $X \perp C$; the event-time and censoring distributions are continuous on $[0, t]$; and the risk-set probability remains positive up to t . We also assume that the asymptotic linear representations in (7) and (8) hold with finite second moments. For regular asymptotically linear estimators, first-order variances and covariances are obtained from second moments and cross-products of their influence functions (Van der Vaart 2000; Cole et al. 2025).

Proposition 1 (Bivariate normal limit). Under the conditions stated above,

$$\sqrt{n} \begin{pmatrix} \hat{S}_{KM}(t) - S(t) \\ \tilde{S}_n(t) - S(t) \end{pmatrix} \xrightarrow{d} \mathcal{N}_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma(t) \right), \quad (12)$$

where

$$\Sigma(t) = \begin{pmatrix} V_{KM}(t) & V_{\tilde{S}}(t) \\ V_{\tilde{S}}(t) & V_{\tilde{S}}(t) \end{pmatrix}, \quad (13)$$

with

$$V_{KM}(t) = E\{\text{IF}_{KM}^2\}, \quad V_{\tilde{S}}(t) = E\{\text{IF}_{\tilde{S}}^2\} = S(t)\{1 - S(t)\}. \quad (14)$$

Equivalently, on the estimator scale,

$$\frac{1}{n} \Sigma(t) = \begin{pmatrix} v_{KM}(t) & v_{\tilde{S}}(t) \\ v_{\tilde{S}}(t) & v_{\tilde{S}}(t) \end{pmatrix}, \quad (15)$$

where $v_{KM}(t) = V_{KM}(t)/n$ and $v_{\tilde{S}}(t) = V_{\tilde{S}}(t)/n$.

Proof. By the asymptotic linear representations in (7) and (8),

$$\sqrt{n} \begin{pmatrix} \widehat{S}_{KM}(t) - S(t) \\ \widetilde{S}_n(t) - S(t) \end{pmatrix} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \begin{pmatrix} \text{IF}_{KM,i} \\ \text{IF}_{\widetilde{S},i} \end{pmatrix} + o_p(1). \quad (16)$$

The summands are iid mean-zero vectors with finite second moments. The multivariate central limit theorem and Slutsky's theorem therefore give the stated bivariate normal limit, with covariance matrix

$$\Sigma(t) = E \left[\begin{pmatrix} \text{IF}_{KM} \\ \text{IF}_{\widetilde{S}} \end{pmatrix} \begin{pmatrix} \text{IF}_{KM} & \text{IF}_{\widetilde{S}} \end{pmatrix} \right]. \quad (17)$$

It remains only to identify the entries of this matrix. For the complete-data empirical estimator,

$$\text{IF}_{\widetilde{S}}(X; t) = I(X > t) - S(t), \quad (18)$$

and hence

$$E\{\text{IF}_{\widetilde{S}}^2\} = \text{var}\{I(X > t)\} = S(t)\{1 - S(t)\} = V_{\widetilde{S}}(t). \quad (19)$$

Using the Peterson–Reid decomposition in (9),

$$\begin{aligned} E\{\text{IF}_{KM} \text{IF}_{\widetilde{S}}\} &= E\{(\text{IF}_{\widetilde{S}} + Z_t) \text{IF}_{\widetilde{S}}\} \\ &= E\{\text{IF}_{\widetilde{S}}^2\} + E\{Z_t \text{IF}_{\widetilde{S}}\} \\ &= V_{\widetilde{S}}(t), \end{aligned} \quad (20)$$

because $E\{Z_t \text{IF}_{\widetilde{S}}\} = 0$. Similarly,

$$\begin{aligned} E\{\text{IF}_{KM}^2\} &= E\{(\text{IF}_{\widetilde{S}} + Z_t)^2\} \\ &= E\{\text{IF}_{\widetilde{S}}^2\} + 2E\{Z_t \text{IF}_{\widetilde{S}}\} + E\{Z_t^2\} \\ &= V_{\widetilde{S}}(t) + \text{var}(Z_t) = V_{KM}(t). \end{aligned} \quad (21)$$

Combining these identities gives

$$\Sigma(t) = \begin{pmatrix} V_{KM}(t) & V_{\widetilde{S}}(t) \\ V_{\widetilde{S}}(t) & V_{\widetilde{S}}(t) \end{pmatrix}. \quad (22)$$

Dividing by n gives the corresponding estimator-scale covariance matrix. $\square$

The covariance identity also has a useful heuristic interpretation in addition to the formal influence-function derivation. To first order, $\widehat{S}_{KM}(t)$ behaves like the complete-data empirical estimator $\widetilde{S}_n(t)$ plus an additional mean-zero censoring noise:

$$\widehat{S}_{KM}(t) \approx \widetilde{S}_n(t) + \text{censoring noise}. \quad (23)$$

The Peterson–Reid decomposition shows that this censoring noise is orthogonal to the complete-data component. Therefore,

$$\begin{aligned} \text{cov}\{\widehat{S}_{KM}(t), \widetilde{S}_n(t)\} &\approx \text{cov}\{\widetilde{S}_n(t) + \text{censoring noise}, \widetilde{S}_n(t)\} \\ &= \text{var}\{\widetilde{S}_n(t)\} = v_{\widetilde{S}}(t). \end{aligned} \quad (24)$$

Thus censoring adds noise to the Kaplan–Meier estimator and inflates its variance, but this added noise is not shared with the latent complete-data empirical estimator due to independent censoring and orthogonality of influence function components. The leading covariance is therefore the latent complete-data variance.

3. Applications and Examples

3.1. Relative Efficiency and Effective Sample Size

The variance decomposition suggests a natural way to measure the amount by which censoring inflates variability. Equation (11) decomposes the Kaplan–Meier variance into a complete-data component, $v_{\widetilde{S}}(t)$, and an additional nonnegative term attributable to censoring. Dividing both sides of (11) by $v_{\widetilde{S}}(t)$ motivates the variance-inflation factor

$$\varphi(t) = \frac{v_{KM}(t)}{v_{\widetilde{S}}(t)} \geq 1, \quad (25)$$

which measures the multiplicative increase in variability—or, equivalently, the proportional loss of information—induced by censoring. When there is no censoring, $v_{KM}(t) = v_{\widetilde{S}}(t)$ and $\varphi(t) = 1$. Intuitively, $\varphi(t)$ measures how much extra “noise” is introduced because some event times remain unobserved, while the underlying sampling process itself is unchanged. When no censoring occurs, $\varphi(t) = 1$ and the Kaplan–Meier estimator coincides with the empirical survival estimator, indicating that all information is retained. This inflation factor parallels the concept of asymptotic relative efficiency (ARE)—the ratio of sample sizes required for two estimators to achieve equal large-sample precision (Serfling 2025). Here, $1/\varphi(t)$ can be interpreted as the pointwise relative efficiency

$$RE(t) = \frac{v_{\widetilde{S}}(t)}{v_{KM}(t)} = \frac{1}{\varphi(t)}, \quad (26)$$

representing the fraction of information retained under censoring (Oakes 1977). Equivalently, define the effective sample size

$$ESS(t) = n RE(t) = \frac{n}{\varphi(t)} = \frac{S(t)\{1 - S(t)\}}{v_{KM}(t)}, \quad (27)$$

that is, the size of a hypothetical complete-follow-up binomial experiment at time t with success probability $S(t)$ (survival beyond t) whose variance $S(t)\{1 - S(t)\}/ESS(t)$ matches the large-sample variance of the Kaplan–Meier estimator at time t . This connects directly with the information-based measures proposed by Benedetti et al. (1982) and revisited for the KM estimator by Hackmann et al. (2025).

For practical use, plug-in estimators based on Greenwood's variance (6) and binomial variance estimator $\widehat{S}_{KM}(1 - \widehat{S}_{KM})n^{-1}$ for the latent survival provide

$$\widehat{RE}(t) = \frac{1 - \widehat{S}_{KM}(t)}{n \widehat{S}_{KM}(t) \widehat{\sigma}^2(t)}, \quad \widehat{ESS}(t) = \frac{1 - \widehat{S}_{KM}(t)}{\widehat{S}_{KM}(t) \widehat{\sigma}^2(t)}. \quad (28)$$

These quantities convert the theoretical variance decomposition into practical, interpretable measures of information retention and precision at each time point.

3.2. Correlation and Concordance under Censoring

Both the Kaplan–Meier estimator $\widehat{S}_{KM}(t)$ and the hypothetical complete-data estimator $\widetilde{S}_n(t)$ are consistent for the true survival function $S(t)$:

$$\widehat{S}_{KM}(t) \xrightarrow{p} S(t), \quad \widetilde{S}_n(t) \xrightarrow{p} S(t). \quad (29)$$

However, in finite samples they differ because censoring contributes a distinct source of variation. While $\tilde{S}_n(t)$ is unobservable, its relationship to $\hat{S}_{KM}(t)$ can be characterized in terms of their covariance structure derived from the influence-function decomposition ((9) and Proposition 1). This result formalizes the intuitive idea that censoring not only inflates variance but also weakens the correlation between observed and hypothetical complete-data estimates.

Proposition 2 (Correlation under censoring). For fixed t , under independent right censoring,

$$\rho(t) := \text{corr}\{\hat{S}_{KM}(t), \tilde{S}_n(t)\} = \sqrt{\frac{v_{\tilde{S}}(t)}{v_{KM}(t)}} = \frac{1}{\sqrt{\varphi(t)}}. \quad (30)$$

Thus, censoring attenuates the correlation between the observed and latent survival estimates in proportion to the variance-inflation factor. When censoring is absent, $\varphi(t) = 1$ and $\rho(t) = 1$, indicating perfect concordance.

Remark 1 (Temporal correlation). In the absence of censoring, survival estimates at two times $t_i < t_j$ have correlation

$$\rho_0(t_i, t_j) = \sqrt{\frac{S(t_j)\{1 - S(t_i)\}}{S(t_i)\{1 - S(t_j)\}}}. \quad (31)$$

Censoring further reduces this correlation approximately by the ratio $\sqrt{\varphi(t_i)/\varphi(t_j)}$, leading to

$$\rho(t_i, t_j) \approx \rho_0(t_i, t_j) \sqrt{\frac{\varphi(t_i)}{\varphi(t_j)}}. \quad (32)$$

In practice, survival probabilities are often summarized or compared at selected fixed time points (Klein et al. 2007) (e.g., one-year survival), yet conclusions are sometimes implicitly extended to other time points. When $\rho(t_i, t_j)$ is low, estimates at different times contain largely independent information, and inference at one time point provides little evidence about another. Hence, fixed-time analyses should recognize that information decays with both elapsed time and accumulated censoring.

Figure 1 illustrates (32) in an exponential survival model with mean survival 365 days, using three fixed-time points $t_1 = 130$, $t_2 = 186$, and $t_3 = 439$ days, so that (t_1, t_2) represents a nearby pair and (t_1, t_3) a widely separated pair on the scale of follow-up. Under no censoring, the corresponding baseline correlations are $\rho_0(t_1, t_2) \approx 0.80$ and $\rho_0(t_1, t_3) \approx 0.43$, reflecting the natural decay of temporal dependence even with complete data. As censoring increases, the observed Kaplan–Meier correlations fall systematically below these baselines, with only mild attenuation for (t_1, t_2) but a much larger reduction for (t_1, t_3) . This produces the clear ordering

$$\sqrt{\varphi(t_1)/\varphi(t_2)} \gg \sqrt{\varphi(t_1)/\varphi(t_3)}, \quad (33)$$

demonstrating that censoring primarily weakens longer-range temporal dependence, while shorter-range dependence is less affected.

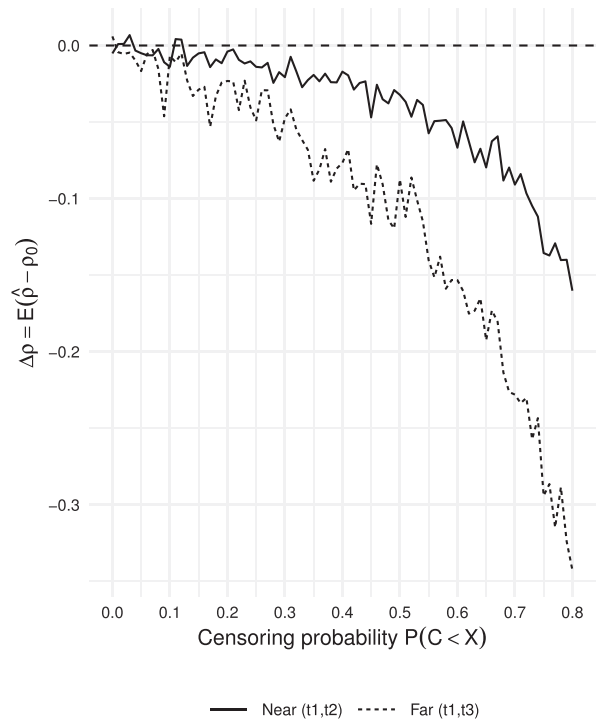


Figure 1. Censoring-induced attenuation of temporal correlation between Kaplan–Meier estimates at fixed time points $t_1 = 130$, $t_2 = 186$, and $t_3 = 439$ days (Near: (t_1, t_2) ; Far: (t_1, t_3)) for exponential survival times with mean survival of 365 days. Curves show Monte Carlo averages over 5000 simulated trials of the correlation gap $\Delta\rho = \hat{\rho}(t_i, t_j) - \rho_0(t_i, t_j)$ relative to the no-censoring reference $\rho_0(t_i, t_j)$; more negative values indicate stronger censoring-induced decrease in correlation.

3.3. Latent-Observed Difference and the Censoring-Component Interval

Define the latent-observed gap

$$D_n(t) = \tilde{S}_n(t) - \hat{S}_{KM}(t). \quad (34)$$

Using Proposition 1 and $\text{cov}\{\hat{S}_{KM}(t), \tilde{S}(t)\} = v_{\tilde{S}}(t)$,

$$\begin{aligned} \text{Var}\{D_n(t)\} &= \text{Var}\{\tilde{S}(t)\} + \text{Var}\{\hat{S}_{KM}(t)\} - 2\text{Cov}\{\hat{S}_{KM}(t), \tilde{S}(t)\} \\ &\approx v_{\tilde{S}}(t) + v_{KM}(t) - 2v_{\tilde{S}}(t) \\ &= v_{KM}(t) - v_{\tilde{S}}(t) \\ &= v_{\tilde{S}}(t)\{\varphi(t) - 1\}. \end{aligned} \quad (35)$$

Define the corresponding censoring-only standard error at time t as

$$\begin{aligned} \text{SE}_{cc}(t) &= \sqrt{v_{KM}(t) - v_{\tilde{S}}(t)} \\ &= \sqrt{v_{\tilde{S}}(t)\{\varphi(t) - 1\}}. \end{aligned} \quad (36)$$

A pointwise $100(1 - \alpha)\%$ censoring-component interval, $\text{CI}_{cc}(t)$, for the latent complete-data estimator $\tilde{S}_n(t)$ is obtained by propagating $\text{SE}_{cc}(t)$ through a standard link transformation. Fix a smooth, strictly monotone link $h : (0, 1) \rightarrow \mathbb{R}$ (Borgan and Liestøl 1990) (e.g., log–log, logit, or arcsine-square-root).

In our implementation we use the log–log transformation as the default link, consistent with standard Kaplan–Meier survival intervals (Borgan and Liestøl 1990; Xiang, Gao, and Jin 1995). This transformation keeps back-transformed limits within $[0, 1]$

and has favorable finite-sample behavior in standard survival settings. The choice of link mainly affects finite-sample shape and boundary behavior. Under the delta method, all smooth monotone links are based on the same variance on the survival scale, but differ in how this variance is transformed and back-transformed into an interval. Thus, different links represent the same censoring-induced uncertainty while allocating interval width differently between the lower and upper tails. As sensitivity analyses, we also consider the logit and arcsine-square-root transformations. Additional simulations comparing the log-log, logit, and arcsine-square-root links are reported in the supplemental material; across the settings considered, the censoring-component interval showed similar pointwise coverage for all three transformations.

The delta-method gives the pointwise censoring-component interval for the latent complete-data estimator $\tilde{S}_n(t)$:

$$CI_{cc}(t) := h^{-1}\left(h\{\widehat{S}_{KM}(t)\} \pm z_{1-\alpha/2} |h'\{\widehat{S}_{KM}(t)\}| SE_{cc}(t)\right). \quad (37)$$

The interval in (37) should not be interpreted as a confidence interval for the population survival probability $S(t)$. Standard Kaplan–Meier intervals target $S(t)$ at a fixed time point and approximately satisfy

$$\Pr\{S(t) \in CI_{KM}(t)\} \approx 1 - \alpha \quad \text{for each fixed } t. \quad (38)$$

In contrast, (37) is a pointwise interval for the unobserved complete-follow-up estimator $\tilde{S}_n(t)$ —the survival proportion that would be obtained from the same sample if everyone were observed through time t (i.e., no censoring up to t); under the working assumptions used to construct (37), $\tilde{S}_n(t)$ falls inside $CI_{cc}(t)$ with probability approximately $1 - \alpha$, that is,

$$\Pr\{\tilde{S}_n(t) \in CI_{cc}(t)\} \approx 1 - \alpha. \quad (39)$$

Because the construction is pointwise, it does not provide simultaneous coverage over all observed event times; that is, for a grid $\{t_1, \dots, t_{N(t)}\}$, in general,

$$\Pr\{\tilde{S}_n(t_j) \in CI_{cc}(t_j) \text{ for all } j = 1, \dots, N(t)\} < 1 - \alpha, \quad (40)$$

and this joint probability typically decreases as the cumulative event count $N(t)$ increases (Nemes 2025). Accordingly, (37) should not be read as an entire-curve band. Rather, one expects that roughly a proportion $1 - \alpha$ of evaluated time points satisfy $\tilde{S}_n(t_j) \in CI_{cc}(t_j)$, with the remaining fraction falling outside (possibly in several separate stretches of follow-up). Finally, in the absence of censoring ($\varphi(t) = 1$), $v_{KM}(t) = v_{\tilde{S}}(t)$ so that $SE_{cc}(t) = 0$ and $CI_{cc}(t)$ collapses to a single point, reflecting $\tilde{S}_n(t) = \widehat{S}_{KM}(t)$. $\widehat{ESS}(t)$ and the censoring-component pointwise interval are implemented using plug-in estimators $\widehat{\varphi}(t)$ and $\widehat{v}_{\tilde{S}}(t)$ computed from standard KM/Greenwood quantities. This is a first-order delta-method construction: under the usual regularity conditions for the Kaplan–Meier estimator, $\widehat{\varphi}(t) \rightarrow \varphi(t)$ and $\widehat{v}_{\tilde{S}}(t) \rightarrow v_{\tilde{S}}(t)$ in probability, and replacing unknown functionals by consistent estimators does not affect the leading $n^{-1/2}$ asymptotic behavior (Slutsky’s theorem). Consequently, the additional variability induced by estimating $\varphi(t)$ and $v_{\tilde{S}}(t)$ influences the interval width only at higher order, while the dominant contribution is the sampling variability of $\widehat{S}_{KM}(t)$ itself.

3.4. Contribution of Censored Observations and the Risk-Set History

To make the role of censoring and the interpretation of $ESS(t)$ more concrete, we briefly revisit the likelihood underlying the Kaplan–Meier estimator. We write the discrete hazard at event time t_j as α_j . Under independent right censoring, the likelihood for the sequence of hazards factorizes as a product of binomial terms (Kaplan and Meier 1958):

$$L(\alpha_1, \dots, \alpha_m) \propto \prod_{j=1}^m \alpha_j^{d_j} \{1 - \alpha_j\}^{R(t_j) - d_j}. \quad (41)$$

At each event time t_j , this represents a binomial experiment with “successes” d_j and “trials” $R(t_j)$. All individuals who are at risk just before t_j contribute to this term, including those who will later be censored. Thus, a subject who is eventually censored at time c contributes one trial to the binomial likelihood at every event time $t_j < c$. Maximizing (41) yields the local estimator $\widehat{\alpha}_j = d_j/R(t_j)$, with familiar binomial variance estimate

$$\widehat{\text{Var}}(\widehat{\alpha}_j) = \frac{\widehat{\alpha}_j\{1 - \widehat{\alpha}_j\}}{R(t_j)}. \quad (42)$$

Conditional on the past, the incremental precision at t_j is therefore determined by the risk-set size $R(t_j)$: adding one more individual at risk (even if censored later) increases $R(t_j)$ and decreases the local variance. Through Greenwood’s formula, these local contributions propagate into the cumulative variance $v_{KM}(t)$. To formalize this dependence, let $\mathcal{F}_t = \sigma\{N(u), N^c(u) : 0 \leq u \leq t\}$ be the filtration generated by the event and censoring processes. Because the estimator aggregates the local precision at every past event time, its variance $v_{KM}(t)$ is an $\mathcal{F}_t$ -measurable functional of the entire risk-set trajectory $\{R(t_j) : t_j \leq t\}$, not just the current snapshot $R(t)$. Two censoring patterns can have the same current number at risk but different past trajectories, and hence different accumulated variances and $ESS(t)$. By definition, $ESS(t)$ is the unique value that equates the binomial variance $S(t)\{1 - S(t)\}/ESS(t)$ to this path-dependent variance $v_{KM}(t)$. Thus, $ESS(t)$ compresses the entire risk-set history up to time t into a single binomial variance-equivalent sample size. In a likelihood sense, censored participants contribute partial information through their earlier follow-up, even though they do not generate events at the tail.

4. Numerical and Real-World Examples

This section empirically illustrates the theoretical results developed above through both simulation and real-data analyses. The simulations in Section 4.1 validate the variance decomposition and information measures introduced earlier, showing that (i) the effective sample size (ESS) captures not only variance but also finite-sample normal-approximation behavior, and (ii) the proposed censoring-only interval maintains nominal coverage across survival probabilities. Section 4.2 applies the same framework to CheckMate-057 and ACTS-CC, demonstrating how ESS and censoring-component intervals quantify information retained under censoring in practice and clarify why small risk sets need not imply weak statistical evidence.

4.1. Simulation Study

The simulation study serves two purposes: first, to validate that the proposed information-based quantities reproduce the finite-sample behavior of the Kaplan–Meier estimator, and second, to demonstrate that the effective sample size, $ESS(t)$, captures history-dependent information loss that the simple number at risk, $R(t)$, ignores.

4.1.1. Validation of Finite-Sample Properties.

We first assess whether the finite-sample properties implied by the variance decomposition hold in practice. Specifically, we evaluated (i) whether replacing the nominal sample size n with $ESS(t)$ reproduces the same normal-approximation accuracy (Berry–Esseen bound) as the Kaplan–Meier estimator, and (ii) whether the censoring-component interval derived from [Proposition 1](#) achieves its nominal coverage probability.

We simulated right-censored data from $X_i \sim \text{Exp}(1)$ and independent $C_i \sim \text{Unif}(0, c_p)$, where c_p was calibrated to yield approximately 50% censoring. For each of 5000 replications ($n = 250$), we computed the Kaplan–Meier estimator $\hat{S}_{KM}(t)$ and its log–log confidence interval, along with the censoring-component interval derived from (37). The effective sample size $ESS(t)$ was used to construct a binomial Berry–Esseen bound as a benchmark for the bound derived for the KM estimator (Chang and Rao 1989). Mean Berry–Esseen bounds and empirical coverage probabilities were evaluated across survival probabilities $S(t) \in [0.2, 0.9]$.

Panel (a) of [Figure 2](#) compares the mean Berry–Esseen bounds for the KM estimator and the binomial benchmark using $ESS(t)$. The near-overlap of the two curves indicates that replacing n with $ESS(t)$ recovers the same finite-sample normal-approximation behavior, confirming that $ESS(t)$ encapsulates both asymptotic variance and finite-sample accuracy. Panel (b) shows empirical coverage of the censoring-component (cc) interval versus the standard log–log Kaplan–Meier confidence

interval. Both maintain coverage close to 95% across the range of survival probabilities, confirming that the additional variance term $v_{\tilde{S}}(t)\{\varphi(t) - 1\}$ correctly accounts for censoring-induced uncertainty.

4.1.2. The Impact of Risk-Set History.

To demonstrate that $ESS(t)$ captures the information lost due to censoring history—which the current number at risk $R(t)$ fails to capture—we conducted a second simulation comparing two scenarios with identical risk-set sizes at the analysis time but different censoring trajectories. We simulated event times from an exponential distribution with rate $\lambda = 1/365$ days (median survival ≈ 253 days) for $n = 1,000$ subjects. Censoring times were generated from Weibull distributions whose scale parameters were numerically calibrated (for each shape) to ensure that 50% of subjects remained uncensored at the median survival time (t_{med}), implying $E\{R(t_{med})\} \approx 250$ in both scenarios, with the scenarios differing only in the censoring shape parameter. The first, Sharp Early Censoring (Weibull $k = 0.2$), followed a “hyper-exponential” pattern where censoring occurred rapidly after enrollment, depleting the risk set and leaving relatively little accumulated follow-up. Conversely, the Late Censoring scenario (Weibull $k = 5.0$) reflected an administrative-like pattern where most subjects remained under observation for the majority of the study, with censoring concentrated shortly before t_{med} . As shown in [Table 1](#), both scenarios resulted in virtually identical numbers at risk at the analysis time, yet the information retained

Table 1. Simulation results comparing information retention under Sharp Early versus Late censoring at the median survival time.

Scenario	$\widehat{R}(t)$	$\widehat{V}_{KM}$	$\widehat{ESS}(t)$	$SD_{mc}(ESS)$
Sharp early (Weibull $k = 0.2$)	250.0	4.60×10^{-4}	543.1	10.9
Late (Weibull $k = 5.0$)	250.1	3.06×10^{-4}	816.9	16.3

NOTE: Despite virtually identical numbers at risk ($R(t)$), the Effective Sample Size ($ESS(t)$) reveals a significant difference in information content.

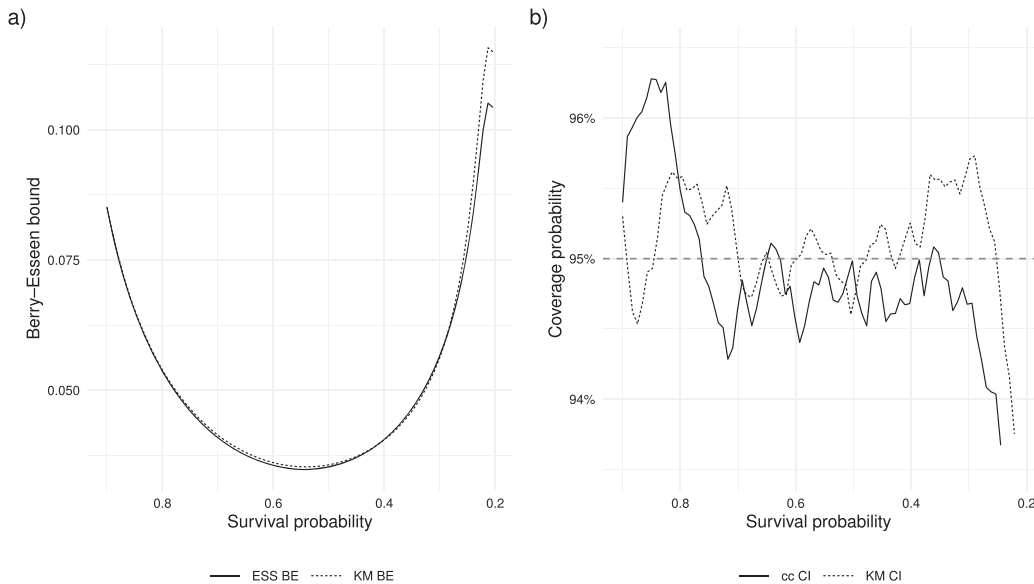


Figure 2. Finite-sample validation of the variance decomposition. (a) Mean Berry–Esseen bounds for the Kaplan–Meier estimator (solid) and for the binomial benchmark based on $ESS(t)$ (dashed). (b) Empirical pointwise coverage of the censoring-component (cc) interval and the standard Kaplan–Meier 95% log–log pointwise confidence interval; both remain close to the nominal 95% level.

differed substantially. The Late Censoring scenario yielded a lower variance than the Sharp Early scenario, resulting in an effective sample size that was approximately 1.50 times larger (816.9 vs. 543.1). This illustrates that while $R(t)$ provides a snapshot of the risk set at time t and does not reflect the preceding censoring trajectory, $\widehat{ESS}(t)$ accounts for the accumulated loss of information over time and accordingly distinguishes the higher precision associated with a more sustained risk-set history.

Together, these results demonstrate that $\widehat{ESS}(t)$ provides an operational link between the theoretical variance decomposition and finite-sample performance. At the same time, $R(t)$ still has a natural forward-looking interpretation: it is the size of the cohort still under observation just before t , and therefore sets an upper bound on the additional information that can be accrued after t and hence the maximal precision attainable for estimands evaluated at horizons $u > t$ given the study history up to t .

Additional simulations reported in supplemental materials extend the evaluation to Weibull and log-normal survival times, with marginal censoring probabilities from 20% to 70%. The results were similar across event-time distributions: the realized risk sets matched their theoretical targets, and the plug-in effective sample size closely tracked the population ESS target. The largest deviations occurred only under heavy censoring at later survival landmarks, where the expected risk set was small. These findings indicate that the main conclusions are not specific to exponential survival times.

4.2. Assessing Information Retention in Clinical Trials

We illustrate the practical implications of the proposed framework using two phase III randomized oncology trials with time-to-event endpoints. The data are available in the `kmdata` R package (Redd, Fell, and Rahman 2022). The first is *CheckMate 057* (Borghaei et al. 2015), which compared nivolumab (3 mg/kg every two weeks) with docetaxel (75 mg/m² every three weeks) in 582 patients with previously treated advanced non-squamous non-small-cell lung cancer; the primary endpoint was overall survival, with one-year outcomes serving as key efficacy benchmarks. The second is the *ACTS-CC* trial (Yoshida et al. 2014), an open-label randomized phase III noninferiority study in 1518 patients with curatively resected stage III colon cancer, comparing adjuvant S-1 with tegafur-uracil plus leucovorin (UFT/LV), with three-year disease-free survival as the primary endpoint. Table 2 reports $\widehat{S}_{KM}(t)$, $R(t)$, and $\widehat{ESS}(t)$ at prespecified fixed times for both trials.

At 12 months in *CheckMate 057*, $\widehat{ESS}(t)$ remained close to the baseline sample size, whereas the number at risk $R(t)$ had

already declined substantially, largely due to observed events rather than censoring. This difference reflects information content: subjects with events contribute essentially complete time-to-event information, while censored subjects contribute only their observed event-free follow-up (on average 2.5 months in the docetaxel arm and 5.2 months in the nivolumab arm prior to 12 months). The trial report (Borghaei et al. 2015) does not attribute the later rise in censoring to a specific cause; it is consistent with administrative right-censoring at the interim data cutoff and limited long-term follow-up, concentrating censoring in the right tail and reducing $R(t)$ to single digits by 24 months. Even so, censored subjects contributed substantial follow-up before censoring: among those censored prior to 24 months, the average event-free follow-up was 15.8 months (docetaxel) and 17.7 months (nivolumab). This late concentration of censoring is also evident in Figure 3(a) through the widening censoring-component intervals, which account for most of the increase in the standard 95% pointwise intervals beyond approximately 18 months.

The *ACTS-CC* trial (Yoshida et al. 2014) exhibits a different pattern. By 36 months, risk sets remain large and $\widehat{ESS}(t)$ remains close to the original sample size, consistent with limited information loss from censoring at the primary time point of interest. Moreover, censoring occurs relatively late among censored participants, with average event-free follow-up of 25.5 months (S-1) and 26.6 months (UFT/LV), implying a comparatively small censoring-related component of uncertainty at 36 months (Figure 3(b)).

5. Discussion

This article provides an overview of how censoring affects the precision of nonparametric survival estimates. By decomposing the variance of the Kaplan–Meier estimator into complete-data and censoring-induced components, we show that the variance-inflation factor, relative efficiency, and effective sample size quantify information retention directly from observable quantities. These results connect classical asymptotic theory with the practical interpretation of survival curves. The decomposition clarifies that variability in the Kaplan–Meier estimator arises from two sources: random sampling and censoring. Only the former exists under full observation; the latter inflates uncertainty in proportion to $\varphi(t)$. This isolates the information loss due to incomplete follow-up without altering the underlying sampling model, extending the full-information benchmark of Gebski et al. (2018) and linking it to influence-function theory.

Table 2. Fixed-time survival summaries for *CheckMate 057* and *ACTS-CC*: Kaplan–Meier estimate $\widehat{S}_{KM}(t)$, number of events $N(t)$, number censored $N^c(t)$, number at risk $R(t)$, plug-in effective sample size $\widehat{ESS}(t)$, and $\widehat{ESS}(t)/\widehat{R}(t)$.

Time	Arm	N	$\widehat{S}(t)$	$N(t)$	$N^c(t)$	$R(t)$	$\widehat{ESS}(t)$	$\widehat{ESS}(t)/\widehat{R}(t)$
<i>CheckMate 057 (overall survival)</i>								
12 months	Nivolumab	292	50.3%	144	4	145	289	1.99
	Docetaxel	290	39.2%	172	8	111	282	2.54
24 months	Nivolumab	292	25.9%	189	94	9	155	17.2
	Docetaxel	290	12.0%	222	63	5	92	18.4
<i>ACTS-CC (disease-free survival)</i>								
36 months	S-1	758	75.6%	180	83	495	734	1.48
	UFT/LV	760	72.4%	204	92	466	735	1.58

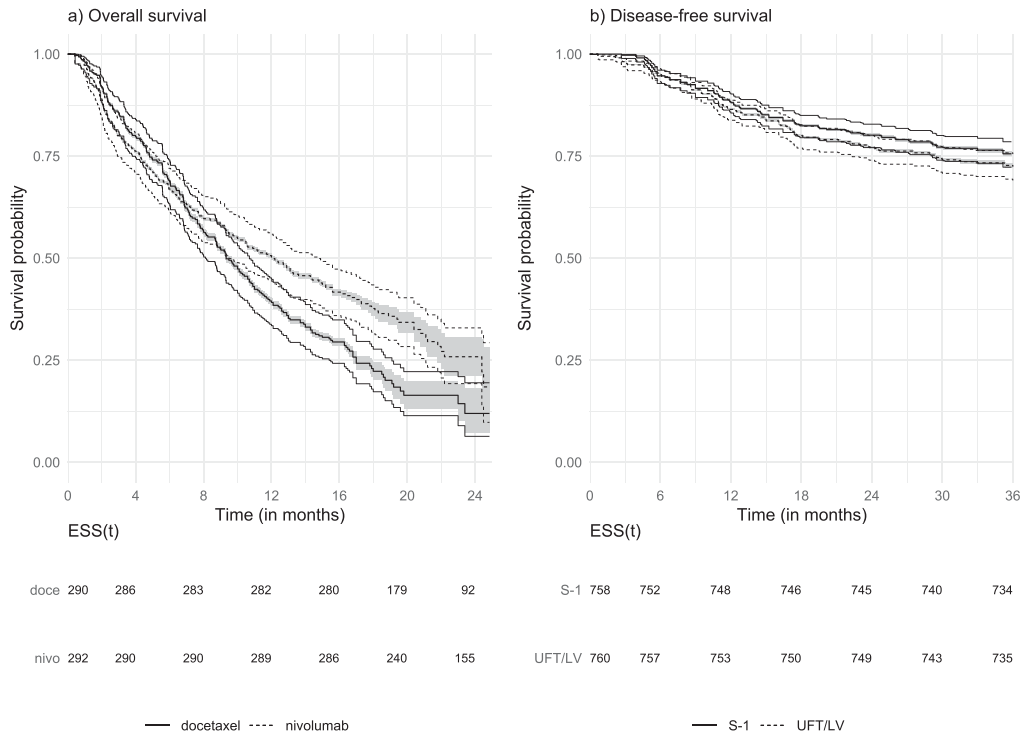


Figure 3. Overall survival in CheckMate 057 (a) and disease-free survival in ACTS-CC (b). Kaplan-Meier estimates are shown with pointwise 95% log-log confidence intervals. The shaded regions show pointwise censoring-component limits for the hypothetical complete-data survival estimates, illustrating the finite-sample uncertainty attributable solely to censoring. Plug-in effective sample size, $\bar{ESS}(t)$, at selected times is shown below each panel.

In applied analyses, $ESS(t)$ formalizes what analysts often understand intuitively: censored participants continue to contribute information through earlier follow-up. Hence, small risk sets do not necessarily imply weak precision. Reporting $ESS(t)$ alongside conventional at-risk counts offers a clearer, variance-based summary of the information available at each time point. The pointwise censoring-component interval shows uncertainty attributable solely to censoring, highlighting where observed and latent survival curves may diverge.

The derivation of the effective sample size $ESS(t)$ relies on the standard Kaplan-Meier estimator and Greenwood's formula for variance, which are piecewise constant and update only at observed event times. This characteristic implies that censoring events occurring between failure times do not immediately affect the estimated variance or $ESS(t)$. Miettinen (2008) argues that the standard practice of “carrying forward” the variance estimate unchanged despite a shrinking risk set overlooks the progressive information loss inherent to censoring, potentially yielding overly optimistic variance estimates in intervals where censoring is heavy but events are rare. Addressing this limitation, Hackmann et al. (2025) propose a “modified effective sample size” based on the variance estimator of Dorey and Korn (1987), which explicitly accounts for information reduction between event times. However, retaining the standard formulation is fundamentally rooted in the likelihood and asymptotic structure of the Kaplan-Meier estimator. Under independent right censoring, the likelihood factorizes into a sequence of independent binomial experiments evaluated exclusively at the observed event times. As formally proved by Meier (1975), the asymptotic validity of Greenwood's variance formula rests precisely on treating the estimator as a product of these conditionally indepen-

dent binomial proportions, where the sample size for each discrete step is exactly the risk set $R(t_j)$. Furthermore, Meier highlights that the Kaplan-Meier estimator acts as a nonparametric maximum likelihood estimator by placing probability mass only at observed event times. Consequently, the survival function is formally estimated only at these discrete points. Updating the variance so that it increases between event times would therefore imply estimating a change in precision for a time interval where the survival probability itself is not updated or estimated. Thus, keeping the standard step-function formulation keeps the variance consistent with the point estimate, reflecting the fact that censored subjects contribute to the information strictly through their presence in the risk set at prior event times.

From a reporting perspective, $\bar{ESS}(t)$ is best viewed as a data-maturity measure for Kaplan-Meier summaries. Existing guidance emphasizes that survival curves should not be interpreted simply because they can be computed far into the tail; they should be curtailed or interpreted cautiously when only a small proportion of participants remains in follow-up (Pocock, Clayton, and Altman 2002), and interpretation should depend on the amount of information available at the time point of interest (GebSKI et al. 2018). The present framework makes this idea explicit through a binomial-equivalent interpretation. At a fixed time t , $\bar{ESS}(t)$ is the size of a hypothetical complete-follow-up binomial experiment with success probability $\hat{S}_{KM}(t)$ whose variance equals the Greenwood variance of the Kaplan-Meier estimator. Thus, $\bar{ESS}(t)$ translates the precision of a censored survival estimate into an equivalent complete-data sample size, while $\bar{ESS}(t)/n$ estimates the fraction of complete-data information retained at time t . In clinical trial reports, we recommend presenting $\bar{ESS}(t)$ at clinically prespecified

landmark times, including the primary analysis time and any fixed-time survival summaries. A minimal reporting set would include $\hat{S}_{KM}(t)$, the standard Kaplan–Meier confidence interval, the number at risk $R(t)$, the cumulative number censored $N^c(t)$, $\widehat{ESS}(t)$, and, when useful, $\widehat{ESS}(t)/n$. The number at risk $R(t)$ describes the cohort still under observation just before t , whereas $\widehat{ESS}(t)$ describes the variance-equivalent amount of complete-follow-up information retained at that time. These quantities answer different questions and should not be expected to agree. A small $R(t)$ with large $\widehat{ESS}(t)$ indicates that much of the precision has already been accumulated through earlier events and the preceding risk-set history. In contrast, a small $\widehat{ESS}(t)/n$ indicates substantial information loss from censoring. We do not propose a universal cutoff for $\widehat{ESS}(t)$ or $\widehat{ESS}(t)/n$, because the acceptable level of information retention depends on the clinical context, the landmark time, the original sample size, and whether the estimate is used descriptively or to support decision-making. However, the binomial-equivalent interpretation gives simple reference points. In the hypothetical complete-follow-up binomial experiment underlying $\widehat{ESS}(t)$, the effective number surviving beyond t is $\widehat{ESS}(t)\hat{S}_{KM}(t)$, and the effective number failing before t is $\widehat{ESS}(t)\{1 - \hat{S}_{KM}(t)\}$. Familiar rules of thumb for binomial large-sample approximations, such as requiring both effective counts to exceed about 5 or 10 (Agresti 2013), can therefore be translated directly to the Kaplan–Meier setting. These values should be used only as informal warning signals, not as formal decision rules. They are useful for identifying late survival estimates for which the binomial-equivalent experiment is small or close to the boundary. Accordingly, late Kaplan–Meier estimates should be interpreted cautiously when censoring is heavy, $\widehat{ESS}(t)$ is small, or either binomial-equivalent count is close to zero. The censoring-component interval should be reported as a diagnostic supplement to the usual Kaplan–Meier confidence interval, not as a replacement. The standard Kaplan–Meier interval targets uncertainty about $S(t)$, whereas the censoring-component interval shows where the complete-follow-up empirical survival estimate from the same sample would plausibly have fallen if censoring had not occurred. It is most useful at prespecified landmark times and in survival-curve displays when late follow-up maturity is clinically important. Together, $\widehat{ESS}(t)$, $\widehat{ESS}(t)/n$, the binomial-equivalent counts, and the censoring-component interval provide transparent reporting diagnostics that distinguish late estimates supported by accumulated risk-set history from tail estimates driven mainly by sparse follow-up.

This framework complements earlier efficiency and information measures (Oakes 1977; Benedetti et al. 1982; Hackmann et al. 2025) and integrates naturally with influence—function representations used in robust and semiparametric inference. It provides a coherent view of information decay over time, linking variance inflation, correlation loss, and effective sample size within a single structure. Ultimately, this work reframes survival estimation around information retention rather than sample attrition. The essential question under censoring is not how many subjects remain, but how much information they contribute. By formalizing this perspective through variance decomposition and censoring—only intervals, we offer a theoretically grounded and practically applicable

framework for interpreting uncertainty in censored survival data.

Supplementary Materials

The supplementary materials contain additional simulation results evaluating the plug-in effective sample size and the pointwise coverage of the censoring-component interval across different event-time distributions, censoring levels, and transformation links. R code implementing the proposed methods is also provided.

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Code Availability Statement

R implementation is provided in the online supplementary materials.

AI Use

Generative AI (ChatGPT) and MS Word were used to refine language and clarity. All outputs were reviewed and revised by the author, who assumes full responsibility for the final content.

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